ing 192 single-threaded Ampere cores, setting an industry record for core count. It offers enhanced performance, scalability, and efficiency compared to its predecessors in the series. The processor ensures consistent performance,

enhanced manageability, and robust security, especially in high-demand, multi-user settings like the cloud. AmpereOne caters to every Cloud Native computational demand, from the most energy-efficient constraints to the grandest customer needs. The enhanced core density and reduced power consumption allow cloud providers to present Arm server instances at competitive prices, potentially attracting broader market interest.
Ampere Computing
https://amperecomputing.com

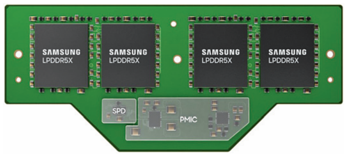
Chip transforming efficient IoT design

Nordic Semiconductor has launched the nRF54H20 multiprotocol system on chip (SoC). The SoC emphasises its potential to lead the way for

advanced IoT solutions once considered out of reach. While most processors prioritise one of these features, the nRF54H20 allows developers to benefit from both by seamlessly switching between configurations. The processor features multiple Arm Cortex-M33 processors and several RISC-V coprocessors. Each is fine-tuned for specific workloads. It is set to disrupt the market, leveraging a potent application processor crafted for intensive processing while optimising power usage. The chip can do many things and uses less energy. Instead of having a standalone general-purpose MCU and a separate wireless SoC, it can all be condensed into one compact SoC. Such integration means future IoT devices will be more energy-efficient, compact, and simpler to develop.
Nordic Semiconductor
https://www.nordicsemi.com

Memory module revolution for PCs and data centres

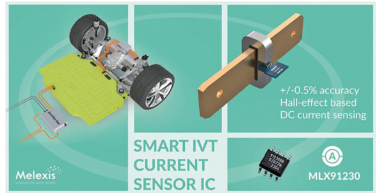
Samsung Electronics has developed the industry's first low-power compression attached memory module (LPCAMM) form factor. This innovation is expected to reshape the



DRAM1 market for PCs, laptops, and possibly even data centres. 7.5 gigabits-per-second (Gbps) LPCAMM has successfully undergone system verification via Intel's platform. LPCAMM bridges the gaps between low-power double data rate (LPDDR) and small-outline dual-in-line memory modules (So-DIMMs), catering to the rising demand for efficient and compact devices. As a detachable module, it offers enhanced flexibility for manufacturers while producing PCs and laptops. It occupies up to 60% less space on the motherboard than So-DIMM, optimising the internal layout of devices while amplifying performance by up to 50% and improving power efficiency by as much as 70%.
Samsung Electronics
https://www.samsung.com/in

Precision sensor transforms EVs and energy storage

Melexis's MLX91230 is the first product of its third generation of current sensors. The compact digital solution provides ±0.5% accuracy cost-effectively. It incorporates integrated voltage and temperature (IVT) measurement, has an embedded microcontroller unit (MCU) to lighten the load on the electronic control unit (ECU), and comes with pre-set safety mechanisms. The sensors suit electric vehicle (EV) battery management and power distribution systems. For original equipment



manufacturers (OEMs) and Tier 1 suppliers aiming to cut costs by internalising DC current sensing design, the sensor offers accurate, reliable monitoring that aligns with top functional safety standards. Unlike other sensors, MLX91230 delivers ±1% accuracy considering thermal drift, lifetime drift, and linearity errors, setting it apart from existing solutions.
Melexis
https://www.melexis.com

Bluetooth module enhances IoT design and features

Murata has launched a Bluetooth Low Energy (LE) module, the Type 2EG, built upon the RSL15 secure Bluetooth

5.2 microcontroller from Onsemi. This certified, secured, and ultra-compact Bluetooth LE module also supports the latest Bluetooth 5.2 features, such as extended range, proximity, high transmission rate, and multi-connection capability, offering the lowest power consumption in its class. It's suitable for IoT devices in the industrial and medical sectors. The Type 2EG stands out for its energy efficiency, owing to its power-savvy radio and built-in microprocessor. This all-in-one, certified solution streamlines IoT device design, accelerating the product launch process.
Murata
https://www.murata.com